

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY (CHARUSAT)
CHARUSAT Campus, Changa – 388 421.

M Sc (MICROBIOLOGY)
SEMESTER III EXAMINATION (BACKLOG)
MI 802: MICROBIAL PHYSIOLOGY

Date: 22.04.2011

Time: 10.00 AM to 1.00 PM

Total Marks: 70

Instructions to the candidates:

1. Provide all necessary information as required in the answer sheet.
2. Attempt all questions.
3. Questions in **Part A** should be answered in the question paper itself. Please **do not write** your candidate ID number on the question paper for Part A.
4. Write answers for Section I and Section II of **Part B** in separate answer sheets.
5. Use of cell phones is strictly prohibited.
6. Use of non-programmable calculators may be allowed if needed.

PART A

Total marks: 20

Q. 1 Choose the correct option and put $\sqrt$ mark in front of it:

- 1 In fermentation pathway NADH is reoxidized by metabolites produced by the pathway.
☐ True ☐ False
- 2 Fructose -6-phospho- phosphoketolase is unique enzyme of the following pathway
(a) Hexose Monophosphate pathway (b) Entner Doudoroff pathway
(c) Bifidum pathway (d) None of the above
- 3 Following organisms contain carboxysomes
(a) Chemolithotrophic sulfur-oxidizing bacteria (b) Cyanobacteria
(c) Nitrifying bacteria (d) All of above
- 4 During fermentation ATP is typically synthesized by
(a) Substrate-level phosphorylation (b) Photophosphorylation
(c) Oxidative phosphorylation (d) None of the above
- 5 Following enzyme is required for assimilatory sulfate reduction.
(a) APS kinase (b) APS reductase (c) Sulfate reductase (d) None of the above
- 6 Following enzyme is coded by *luxD* gene
(a) Acyl-transferase (b) Reductase
(c) Synthetase (d) None of the above
- 7 Resistance to high osmotic lyses is governed by following in mycoplasma.
(a) Sterol (b) Diaminopimelic acid (c) Muramic acid (d) none of the above
- 8 In hydrogen oxidation soluble hydrogenase assimilates H_2
☐ True ☐ False
- 9 The pressure executed by cytoplasmic membrane on the cell wall of bacteria is termed as:
(a) Vapor pressure (b) Turgor pressure (c) Water pressure (d) None of the above

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- 10 DnaJ-DnaK-GrpE team plays an important role in regulation of following stress.
(a) Thermal (b) Osmotic (c) Oxidative (d) All of the above
- 11 The *Fnr* protein in its dimeric form is usually functionally active.
☐ True ☐ False
- 12 Rel A enzyme is essential for the synthesis of pppGpp via the ribosome dependent pathway.
☐ True ☐ False
- 13 Cell division in rod shaped bacteria involves in
(a) Growth of the envelope layers
(b) A septum between two replicated chromosomes Osmotic
(c) Both (d) None of the above
- 14 *min* system, so called because mutants give rise to minicells through
(a) Inappropriate division at the cell poles
(b) Appropriate division at the cell poles
(c) Inappropriate division at the cell center
(d) Appropriate division at the cell center
- 15 *ftsZ* mutants shows no signs of septation, as it acts
(a) Early in the sequence (b) Late in the sequence
(c) Both (d) None of the above
- 16 What is the primary function of endospore formation
(a) Survival of species (b) Dissemination of species
(c) Both (d) None of the above
- 17 Who proposed the concept of cloning DNA directly from environmental samples?
(a) Jo Handelsman (b) Norman Pace (c) Craig Ventor (d) None of the above
- 18 The gene 2135 is defective in yeast cell. While the gene still produces mRNA the defect prevents its translation into protein. Therefore, this gene will display a green spot in the microarray.
☐ True ☐ False
- 19 Which type of genomics studies the physical nature of genomes?
(a) Comparative genomics (b) Structural genomics
(c) Functional genomics (d) None of the above
- 20 The following is a phototrophic nitrogen fixing organism.
(a) *Cyanobacteria* (b) *Azotobacter* (c) *Klebsiella* (d) *Pseudomonas*

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Date: 22.04.2011

Time: 10.00 AM to 1.00 PM

Total Marks: 70

Instructions to the candidates:

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2. Attempt all questions.
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Part B

Total Marks: 50

Section I

- Q.1 (A)** Write a brief note on the following (**any two**) (6)
- (i) Bioluminescence in bacteria
 - (ii) Bifidium pathway and its uniqueness
 - (iii) Lactose fermentation pathway
- Q.1 (B)** Write note on any one of the following (4)
- (i) Methanotrophy
 - (ii) Sulfur oxidation and ATP synthesis
- Q.2 (A)** Write any one of the following (4)
- (i) Electron carrier
 - (ii) Bioenergetics and law of thermodynamics
- Q.2 (B)** Explain briefly (**any two**) (6)
- (i) Oxygenic Photosynthesis
 - (ii) ATP synthesis
 - (iii) Branched ETC and its significance

Section II

- Q.3 (A)** Describe the process of oxidative stress response in *E. coli*. (4)
- Or**
- Q.3 (A)** Explain the function of *Fnr* protein and *ArcAB* system towards nitrate response. (4)
- Q.3 (B)** Write note on **any two** (6)
- (i) Thermal stress and regulation of stress response.
 - (ii) What is ATR? Write in context to *H. pylori*.
 - (iii) Stress response to high osmolality
- Q.4 (A)** Write any one of the following (4)
- (i) Give an account of the steps involved in the gram –ve cocci cell division
 - (ii) ZipA of prokaryotes
- Q.4 (B)** Explain briefly (**any two**) (6)
- (i) Characteristic of the endospore forming bacteria
 - (ii) Graphical representation of the cell fate regulators in *Caulobacter* and the genetic circuit they constitute.
 - (iii) Endospore forming bacteria and disease

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Q.5 (A) Write short notes on: (any two) (6)

- (i) Metagenomics and its importance
- (ii) Nitrogenase complex
- (iii) “cDNA preparation is very important when you are working with eukaryotes” explain.

Q.5 (B) Give brief note on following (any one) (4)

- (i) Metabolic engineering/physiological modifications and its application in current era.
- (ii) DNA microarray as a tool for the Molecular biology.